Part B: Manual Tracing

1. Aim: -

To manually trace and analyze the working of FIFO, LRU, and Optimal Page Replacement Algorithms using a given page reference string and three memory frames.

2. Given Input: -

- Reference String: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2
- Number of Frames: 3
- Total Page References: 13

3. Page -Replacement Algorithms: -

A. FIFO (First-In, First-Out)

FIFO is a page replacement algorithm that replaces the page that entered memory first. It follows the Queue (First-In, First-Out) principle.

When all frames are full and a new page is requested, the oldest page is removed and the new page is inserted.

Working: -

1. Initially, all frames are empty.
2. Pages are loaded into available frames.
3. If a requested page is already present, it is a Page Hit (H).
4. If a requested page is absent, it is a Page Fault (F).
5. When frames are full, the oldest page is replaced.

FIFO Manual Tracing: -

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
Frame 1	7	7	7	2	2	2	2	4	4	4	0	0	0
Frame 2	-	0	0	0	0	3	3	3	2	2	2	2	2
Frame 3	-	-	1	1	1	1	0	0	0	3	3	3	3
F/H	F	F	F	F	H	F	F	F	F	F	F	H	H

FIFO replaces pages in their arrival order, irrespective of how recently they were accessed.

FIFO Result:

- Total Page Faults = 10
- Total Page Hits = 3

B. LRU (Least Recently Used)

LRU is a page replacement algorithm that replaces the page that has not been used for the longest period of time.

It considers the past usage of pages to decide which page should be removed from memory.

Working: -

1. Initially, all frames are empty.
2. Pages are loaded into available frames.
3. If a requested page is already present, it is marked as a Page Hit (H).
4. If a requested page is absent, it is marked as a Page Fault (F).
5. When all frames are full, the page that was used least recently is replaced.
6. Every page hit updates the recent usage of that page.

LRU Manual Tracing: -

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
Frame 1	7	7	7	2	2	2	2	4	4	4	0	0	0
Frame 2	-	0	0	0	0	0	0	0	0	0	3	3	3
Frame 3	-	-	1	1	1	3	3	3	2	2	2	2	2
F/H	F	F	F	F	H	F	H	F	F	F	F	H	H

Explanation of replacement: -

At Step 8, page 4 arrives and all three frames are full.

Current pages are 2, 0 and 3.

- Page 2 was last used at Step 4.
- Page 0 was last used at Step 7.
- Page 3 was last used at Step 6.

Therefore, page 2 is replaced because it was used least recently.

LRU Result: -

- Total Page Faults = 9
- Total Page Hits = 4

C. Optimal Page Replacement: -

Optimal Page Replacement is an algorithm that replaces the page whose next use is farthest in the future.

It examines upcoming page references and selects the page that will be needed last or will not be used again.

Working: -

1. Initially, all frames are empty.
2. Pages are loaded into available frames.
3. If the requested page is already present, it is marked as a Page Hit (H).
4. If the requested page is absent, it is marked as a Page Fault (F).
5. When all frames are full, future page references are checked.
6. The page that will be used farthest in the future is replaced.
7. If multiple pages will never be used again, the first one in the frames is replaced, as specified in the assignment.

Optimal Manual Tracing: -

Ref	7	0	1	2	0	3	0	4	2	3	0	3	2
Frame 1	7	7	7	2	2	2	2	2	2	2	2	2	2
Frame 2	-	0	0	0	0	0	0	4	4	4	0	0	0

Frame 3	-	-	1	1	1	3	3	3	3	3	3	3	3
F/H	F	F	F	F	H	F	H	F	H	H	F	H	H

Explanation of replacement: -

At Step 8, page 4 arrives and the current frames contain 2, 0 and 3.

Future reference string: 2, 3, 0, 3, 2.

- Page 2 will be used first.
- Page 3 will be used next.
- Page 0 will be used after that.

Therefore, page 0 is replaced because its next use is farthest in the future.

Optimal Result: -

- Total Page Faults = 7
- Total Page Hits = 6

Algorithm	Page-Faults	Page-Hits	Hit-Ratio
FIFO	10	3	23.08%
LRU	9	4	30.77%
Optimal	7	6	46.15%

Formula: -

Hit Ratio = Page Hits / Total Page References * 100

Total page references = 13.

Conclusion: -

Optimal Page Replacement performs best because it has the minimum page faults (7) and maximum page hits (6).